

THE EI-REFLECTION OF A CATEGORY, ITS PREORDER REFLECTION, AND THE HOMOMORPHISM ORDER OF GRAPHS

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ABSTRACT. Antisymmetry of an order — $x \leq y$ and $y \leq x$ imply $x = y$ — becomes, when equality is replaced by a class W of morphisms declared to be “sameness”, the condition that every morphism admitting a morphism back lies in W . The morphisms admitting a morphism back form a class $W_{\text{cyc}}(C)$ of any category C , closed under composition and two-out-of-three; it is the largest such class inverted by the preorder reflection, it is $\{\text{id}\}$ exactly for acyclic categories and $\text{Iso}(C)$ exactly for EI-categories, and its localization $C \rightarrow C[W_{\text{cyc}}^{-1}]$ is the reflection of C into EI-categories. The main theorem is that when C has binary coproducts (or binary products) this localization is a preorder: the EI-reflection of C is its preorder reflection $X \preceq Y \iff \text{Hom}(X, Y) \neq \emptyset$. For the topos of graphs this preorder is the homomorphism preorder of Hell and Nešetřil, and on finite graphs the resulting identification is the passage to cores. We contrast the localization with the reflection: in any topos the objects local for W_{cyc} are the subterminal ones, so the saturation of W_{cyc} is the class of maps with equal support, i.e. the (-1) -truncation; W_{cyc} is saturated for sets and simplicial sets, where it is the (-1) -truncated equivalences, but not for graphs. Finally the classes W_{cyc} , W_{obj} (source and target isomorphic), W_{end} (identities and endomorphisms), Aut , Iso are shown to be *natural* — preserved by all functors — and every natural such class other than the class of all morphisms is contained in W_{cyc} . All proofs are elementary and no result depends on a computation.

1. INTRODUCTION

An ordered set is a preorder in which mutual comparability forces equality. Replacing that equality by an isomorphism gives the familiar notion of a skeletal category; replacing it by an identification with more structure leads to univalent categories and complete Segal spaces. This paper studies the intermediate, and elementary, situation in which the equality is replaced by an arbitrary class W of morphisms declared to be a *sameness*: a class

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containing the identities, closed under composition and satisfying two-out-of-three. Such classes form a complete lattice $\text{Same}(C)$ [8], and the question “for which W is C antisymmetric up to W ?” has a canonical answer.

Definition 1.1. For a category C write $X \preceq Y$ if $\text{Hom}(X, Y) \neq \emptyset$ (the *reachability preorder*). A morphism $f: X \rightarrow Y$ has a *back* if $Y \preceq X$. Let

$$W_{\text{cyc}}(C) := \{f: X \rightarrow Y : \text{Hom}(Y, X) \neq \emptyset\}$$

be the class of morphisms with a back (equivalently, the morphisms inside the strongly connected classes of $\preceq$). For $W \in \text{Same}(C)$, the category C is *W-antisymmetric* if $W_{\text{cyc}}(C) \subseteq W$.

Thus a preorder is $\{\text{id}\}$ -antisymmetric iff it is a partial order, and a category is Iso-antisymmetric iff mutually related objects are isomorphic through the relating morphisms. The results are as follows.

The cycle class and the EI-reflection. $W_{\text{cyc}}(C)$ is a sameness, and it is the largest sameness inverted by the preorder reflection $C \rightarrow (\text{Ob}C, \preceq)$ (Theorem 3.1). It is $\{\text{id}\}$ iff C is acyclic, $\text{Iso}(C)$ iff C is an EI-category (all endomorphisms invertible), and $\text{Mor}(C)$ iff C is a coproduct of strongly connected categories (Proposition 3.2). The first new result is that the localization $C \rightarrow C[W_{\text{cyc}}^{-1}]$ is an EI-category and is the *reflection* of C into EI-categories (Theorem 3.3): inverting every morphism that has a morphism back produces the universal EI-category under C . It is standard that in an EI-category the reachability preorder on isomorphism classes of objects is a partial order [6, §9]; Theorem 3.3 goes in the other direction, from an arbitrary category to an EI-category, and identifies the class of morphisms that must be inverted.

Main theorem. The second new result is that if C has binary coproducts, or binary products, then $C[W_{\text{cyc}}^{-1}]$ is a preorder, namely $(\text{Ob}C, \preceq)$ (Theorem 4.1). So for such categories the EI-reflection and the preorder reflection coincide, an EI-category with binary coproducts is a preorder, and the antisymmetrization of the topos of directed graphs is the homomorphism order of graphs [4]: identifying graphs that map to each other is the same as passing to cores (Corollary 4.5).

Localization versus reflection. In a topos the objects local for W_{cyc} are exactly the subterminal ones, so the saturation of W_{cyc} is the class of maps with equal support, the class inverted by the (-1) -truncation (Theorem 6.1). Hence “antisymmetry up to W_{cyc} ” has two shadows: the reflection sees only supports, the localization sees reachability. They agree, and W_{cyc} is saturated, for sets and for simplicial sets — where W_{cyc} is the class of (-1) -truncated equivalences of [9] — but not for graphs.

Natural samenesses. The third new result concerns samenesses defined uniformly for all categories. The classes $\{\text{id}\}$, Aut , Iso , W_{end} (identities and endomorphisms), W_{obj} (source isomorphic to target), W_{ret} (source and

target mutual retracts), W_{cyc} , Mor are samenesses preserved by every functor; they form a lattice, in which Iso and W_{end} are incomparable with join W_{obj} ; and every natural sameness other than Mor is contained in W_{cyc} (Theorem 5.4), so that W_{cyc} is the largest proper natural sameness: a single constraint on every two-out-of-three class that is defined naturally in the category. No result of this paper depends on a computation; small cases were explored by computer only to find the statements.

Related work. EI-categories appear in transformation groups [6] and in the representation theory of categories [11]; we have not seen the observation that they form a reflective subcategory of **Cat** with the localization at W_{cyc} as unit. The preorder and partial-order reflections of a category are standard constructions; the point here is the intermediate EI-reflection, the class of morphisms it inverts, and the fact that under (co)products the two reflections coincide. Categories with a chosen class of “weak equivalences” are the relative categories of Barwick and Kan [1] and the homotopical categories of [2]; samenesses are relative categories whose class satisfies two-out-of-three, and the lattice $\text{Same}(C)$ is the fibre over C of the forgetful functor from relative categories to categories (Remark 2.2). The homomorphism order of graphs and the theory of cores are classical [4]. Subterminal objects, supports and the (-1) -truncation are standard topos theory [5, 7]. The companion papers [8, 9, 10] introduce the sameness lattice and study its sheaf-theoretic and homotopical elements; the present paper is self-contained.

2. PRELIMINARIES

2.1. Conventions. Categories are small unless said otherwise (a topos is a large category, and for it the statements below are applied objectwise or in a larger universe; nothing depends on this). **Cat** denotes the category of small categories.

2.2. Samenesses. A *sameness* on a category C is a class W of morphisms with (i) $\text{id}_X \in W$ for all X , (ii) $g \circ f \in W$ whenever $f, g \in W$ are composable, (iii) two-out-of-three: for composable f, g , if two of $f, g, g \circ f$ lie in W then so does the third. $\text{Same}(C)$ is the set of samenesses ordered by inclusion; it is closed under arbitrary intersections and contains $\text{Mor}(C)$, hence is a complete lattice with $\perp = \{\text{id}\}$ and $\top = \text{Mor}(C)$.

Lemma 2.1. *For a functor $F: C \rightarrow D$ and $V \in \text{Same}(D)$, $F^*(V) := \{f : F(f) \in V\}$ is a sameness on C . In particular $F^*(\text{Iso}(D))$ is the largest sameness inverted by F .*

Proof. F preserves identities and composites, so the three conditions for V transfer to $F^*(V)$. Any class inverted by F is contained in $F^*(\text{Iso}(D))$, which is itself inverted. $\square$

2.3. Localization. For $W \subseteq \text{Mor}(C)$ the localization $\gamma: C \rightarrow C[W^{-1}]$ is the initial functor inverting W [3, I.1]. Its morphisms are represented by zigzags $X \rightarrow A_1 \leftarrow A_2 \rightarrow \cdots \rightarrow Y$ in which the backward arrows lie in W , modulo the relations generated by composition and $w \circ w^{-1} = \text{id}$; in particular $\text{Hom}_{C[W^{-1}]}(X, Y) \neq \emptyset$ iff such a zigzag exists. Note that γ is bijective on objects and that $C[W^{-1}]$ is small when C is.

2.4. Local objects and saturation. For $W \in \text{Same}(C)$ an object Z is W -local if $\text{Hom}(f, Z): \text{Hom}(Y, Z) \rightarrow \text{Hom}(X, Z)$ is bijective for every $f: X \rightarrow Y$ in W ; $\text{Loc}(W)$ denotes the class of W -local objects and $\text{Sat}(W) := \{f: \text{Hom}(f, Z) \text{ bijective for all } Z \in \text{Loc}(W)\}$ the *saturation* of W , a sameness containing W . If $L: C \rightarrow C$ is a reflector onto a full replete subcategory $\mathcal{E}$ then the class $W_L = \{f: L(f) \text{ iso}\}$ is saturated with $\text{Loc}(W_L) = \mathcal{E}$: a map is inverted by L iff $\text{Hom}(f, Z)$ is bijective for all $Z \in \mathcal{E}$, and a W_L -local Z has $\text{Hom}(\eta_Z, Z)$ bijective, whence $\eta_Z: Z \rightarrow LZ$ is a split monomorphism with a retraction r , and $\eta_Z r = \text{id}_{LZ}$ because $\text{Hom}(\eta_Z, LZ)$ is injective; so $Z \cong LZ \in \mathcal{E}$.

Remark 2.2 (relative categories). A pair (C, W) with W a wide subcategory is a relative category [1]; requiring two-out-of-three gives a full subcategory $\mathbf{RelCat}_{2/3}$. The forgetful functor $\mathbf{RelCat}_{2/3} \rightarrow \mathbf{Cat}$ is a bifibration whose fibre over C is $\text{Same}(C)$, with cartesian lifts F^* and cocartesian lifts $F_!$ (the sameness generated by the image); the functor $\text{Same}: \mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$ of [8] is its Grothendieck construction. Nothing below depends on this remark.

3. THE CYCLE CLASS

Theorem 3.1. *For every category C , $W_{\text{cyc}}(C)$ is a sameness, and $W_{\text{cyc}}(C) = P^*(\text{Iso})$ for the preorder reflection $P: C \rightarrow (\text{Ob}C, \preceq)$. Hence $W_{\text{cyc}}(C)$ is the largest sameness inverted by the preorder (and by the partial-order) reflection of C , and*

$$\{\text{id}\} \subseteq \text{Iso}(C) \subseteq W_{\text{obj}}(C) \subseteq W_{\text{cyc}}(C) \subseteq \text{Mor}(C),$$

where $W_{\text{obj}}(C) := \{f: X \rightarrow Y : X \cong Y\}$.

Proof. A morphism of the preorder $(\text{Ob}C, \preceq)$ is invertible iff its endpoints are mutually related, so $P(f)$ is an isomorphism iff f has a back: $W_{\text{cyc}}(C) = P^*(\text{Iso})$, a sameness by Lemma 2.1, and the largest one inverted by P . The partial-order reflection is P followed by the skeleton, which inverts the same morphisms. Identities and isomorphisms have backs; if $X \cong Y$ then any $f: X \rightarrow Y$ has a back; so the chain holds (W_{obj} is a sameness by Lemma 5.2). $\square$

Directly: if $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ have backs h, k then gf has the back hk ; if gf has a back m and f has a back h then g has the back fm ; if gf has a back m and g has a back k then f has the back mg . Figure 1 shows the three cases.

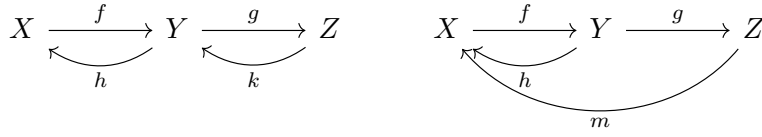


FIGURE 1. Two-out-of-three for W_{cyc} : backs compose (left); if gf and f have backs m and h , then $f \circ m$ is a back for g (right); the third case is symmetric.

Alt text: Two diagrams of three objects in a row with forward arrows f and g ; curved arrows return from Y to X and from Z to Y in the first, and from Y to X and from Z to X in the second.

Proposition 3.2 (the three extremes). *For a category C :*

- (1) $W_{\text{cyc}}(C) = \{\text{id}\}$ iff C is acyclic: $X \preceq Y \preceq X$ implies $X = Y$, and the only endomorphisms are identities;
- (2) $W_{\text{cyc}}(C) = \text{Iso}(C)$ iff C is an EI-category: every endomorphism is invertible;
- (3) $W_{\text{cyc}}(C) = \text{Mor}(C)$ iff C is a coproduct of strongly connected categories, i.e. $\preceq$ is symmetric.

Proof. (1) and (3) are restatements. (2) If C is EI and $f: X \rightarrow Y$ has a back g , then $gf \in \text{End}(X)$ and $fg \in \text{End}(Y)$ are invertible, so f has the left inverse $(gf)^{-1}g$ and the right inverse $g(fg)^{-1}$, which then coincide: f is invertible. Conversely every endomorphism has the identity as a back, so $W_{\text{cyc}}(C) = \text{Iso}(C)$ forces endomorphisms to be invertible. $\square$

Theorem 3.3 (EI-reflection). *For every small category C the localization $C[W_{\text{cyc}}^{-1}]$ is an EI-category with the same reachability preorder as C , and $\gamma: C \rightarrow C[W_{\text{cyc}}^{-1}]$ is universal among functors from C to EI-categories. Hence the full subcategory of (small) EI-categories is reflective in **Cat**, with unit the localization at W_{cyc} .*

Proof. Convexity. If $X \preceq A \preceq Y$ and $Y \preceq X$ then A is in the $\preceq$ -class of X . Consequently every zigzag in C between two objects of one class, with backward arrows in W_{cyc} , stays inside that class: a backward arrow $A_{i+1} \rightarrow A_i$ in W_{cyc} has a back, so $A_i \preceq A_{i+1}$ as well, and inductively all A_i lie between X and Y .

Reachability. A zigzag from X to Y with backward arrows in W_{cyc} yields $X \preceq Y$ by replacing each backward arrow by a back; conversely a morphism $X \rightarrow Y$ is a zigzag. So $\text{Hom}_{C[W_{\text{cyc}}^{-1}]}(X, Y) \neq \emptyset$ iff $X \preceq Y$.

EI. An endomorphism of X in $C[W_{\text{cyc}}^{-1}]$ is represented by a zigzag from X to X , which by convexity lies in the class of X ; every C -morphism between objects of that class has a back, hence is inverted by γ ; so the zigzag is a composite of isomorphisms and their inverses, and is invertible.

Universality. Let $G: C \rightarrow E$ with E an EI-category. If $f: X \rightarrow Y$ has a back g , then $G(f)$ has the back $G(g)$, so $G(f)$ is invertible by Proposition 3.2(2). Thus G inverts W_{cyc} and factors uniquely through γ . $\square$

Remark 3.4. For $W \supseteq W_{\text{cyc}}$ the further localization $C[W^{-1}]$ need not be an EI-category: in the poset with elements $a < b$, $a < d$, $c < b$, $c < d$ and $W = \{a < b, c < d\}$ (so $W \supseteq W_{\text{cyc}} = \{\text{id}\}$), the zigzag $a \rightarrow d \leftarrow c \rightarrow b \leftarrow a$ is an endomorphism of a in $C[W^{-1}]$ which is not invertible, as the localization of this poset at the two edges is the free category on a loop at a together with the remaining arrows. W -antisymmetry is a property of the pair (C, W) , not of the localization.

4. MAIN THEOREM: CATEGORIES WITH BINARY COPRODUCTS

Theorem 4.1. *Let C be a category with binary coproducts, or with binary products. Then $C[W_{\text{cyc}}^{-1}]$ is a preorder: it is the reachability preorder $(\text{Ob}C, \preceq)$. Consequently the EI-reflection of C is its preorder reflection, and the skeleton of $C[W_{\text{cyc}}^{-1}]$ is the partial-order reflection of C .*

Proof. Let $g, g': B \rightarrow C'$ be parallel morphisms of C , and let $j_1, j_2: B \rightarrow B \sqcup B$ be the coproduct injections and $\nabla: B \sqcup B \rightarrow B$ the fold map, so that $\nabla j_1 = \nabla j_2 = \text{id}_B$. The morphism ∇ has the back j_1 , so $\nabla \in W_{\text{cyc}}$ is inverted in $C[W_{\text{cyc}}^{-1}]$, where therefore $j_1 = \nabla^{-1} = j_2$. Hence

$$g = [g, g'] \circ j_1 = [g, g'] \circ j_2 = g' \quad \text{in } C[W_{\text{cyc}}^{-1}],$$

where $[g, g']: B \sqcup B \rightarrow C'$ is the copairing. So any two parallel morphisms of C become equal, and then any two parallel zigzags become equal as well (a zigzag is a composite of morphisms of C and inverses of morphisms of C , and inverses of equal morphisms are equal); thus $C[W_{\text{cyc}}^{-1}]$ has at most one morphism between any two objects. By Theorem 3.3 it has a morphism $X \rightarrow Y$ iff $X \preceq Y$. With binary products use instead the projections $p_1, p_2: C' \times C' \rightarrow C'$, which have the common back Δ , so $p_1 = \Delta^{-1} = p_2$ and $g = p_1 \langle g, g' \rangle = p_2 \langle g, g' \rangle = g'$ (Figure 2). $\square$

$$\begin{array}{c}
 \nabla \\
 \curvearrowleft \\
 B \begin{array}{c} \xrightarrow{j_1} \\ \xrightarrow{j_2} \end{array} B \sqcup B \xrightarrow{[g, g']} C'
 \end{array}$$

FIGURE 2. Theorem 4.1: ∇ has the back j_1 and is inverted; both injections then equal ∇^{-1} , and $g = [g, g']j_1$, $g' = [g, g']j_2$ coincide.

Alt text: An object B with two parallel arrows into the coproduct of B with itself, a curved fold arrow back to B , and the copairing arrow from the coproduct to C' .

Example 4.2 (the hypothesis is needed). Let C be the free category on the graph $x \rightrightarrows y$. Every non-identity morphism has a back, so $W_{\text{cyc}} = \text{Mor}(C)$ and $C[W_{\text{cyc}}^{-1}]$ is the free groupoid on the graph, whose endomorphism group at x is $\mathbb{Z}$. It is an EI-category but not a preorder.

Corollary 4.3. *An EI-category with binary coproducts (or binary products) is a preorder; in particular a groupoid with binary coproducts is equivalent to a discrete category, and a monoid with binary coproducts is trivial.*

Proof. In an EI-category the fold map ∇ , having the back j_1 , is already invertible in C (Proposition 3.2), so the computation in the proof of Theorem 4.1 takes place in C : $j_1 = \nabla^{-1} = j_2$ and $g = g'$ for all parallel g, g' . A groupoid and a monoid are EI-categories. $\square$

Corollary 4.4 (toposes and presheaf categories). *For a topos $\mathcal{E}$, and in particular for every presheaf category $\text{Presh}(S)$, the EI-reflection $\mathcal{E}[W_{\text{cyc}}^{-1}]$ is the hom-preorder of $\mathcal{E}$: $X \preceq Y$ iff there is a morphism $X \rightarrow Y$.*

Corollary 4.5 (graphs and cores). *Let $\mathbf{Gph}$ be the category of directed graphs (presheaves on the category with two objects and two parallel non-identity arrows). Then $\mathbf{Gph}[W_{\text{cyc}}^{-1}]$ is the homomorphism preorder of graphs, $X \preceq Y$ iff there is a graph homomorphism $X \rightarrow Y$; its partial-order reflection is the homomorphism order of [4]. Two finite graphs are identified in $\mathbf{Gph}[W_{\text{cyc}}^{-1}]$ iff they have isomorphic cores.*

Proof. The first statement is Corollary 4.4. Finite graphs X, Y are mutually reachable iff they are homomorphically equivalent, iff their cores are isomorphic [4, Ch. 1]. $\square$

So, for finite graphs, “antisymmetry up to isomorphism” is exactly the passage from a graph to its core, and the sameness lattice records this passage as the single element $W_{\text{cyc}} \in \text{Same}(\mathbf{Gph})$.

5. NATURAL SAMENESSES

Definition 5.1. A *natural sameness* is an assignment $C \mapsto W(C) \in \text{Same}(C)$, for all small categories C , such that $F(W(C)) \subseteq W(D)$ for every functor $F: C \rightarrow D$; equivalently, a subfunctor of the functor $\text{Mor}: \mathbf{Cat} \rightarrow \mathbf{Set}$ that is pointwise a sameness.

Natural samenesses are closed under pointwise intersection, and under pointwise join in $\text{Same}(C)$ (the image of a join is contained in the join of the images, because the sameness generated by $F(W \cup W')$ contains $F(W \vee W')$: the class $F^*(W(D) \vee W'(D))$ is a sameness containing $W(C)$ and $W'(C)$). So they form a lattice $\mathbf{NatSame}$ mapping into every $\text{Same}(C)$ by lattice homomorphisms.

Lemma 5.2. *Let R be an equivalence relation on the objects of C . Then $W_R := \{f: X \rightarrow Y : X R Y\}$ is a sameness (object-determined); if R is preserved by all functors then W_R is natural. Equality gives $\perp$, isomorphism gives W_{obj} , mutual reachability gives W_{cyc} .*

Proof. Reflexivity gives the identities, transitivity gives closure under composition, and transitivity with symmetry gives two-out-of-three ($X R Z$ and $X R Y$ imply $Y R Z$, and similarly in the other cases). $\square$

Proposition 5.3. *The following are natural samenesses:*

- $\perp = \{\text{id}\}$, Iso , W_{obj} (source isomorphic to target), W_{cyc} , $\top = \text{Mor}$;
- $W_{\text{end}}(C) := \{\text{id}\} \cup \{\text{endomorphisms of } C\}$ and $\text{Aut}(C) := \{\text{id}\} \cup \{\text{automorphisms}\} = \text{Iso} \cap W_{\text{end}}$;
- $W_{\text{ret}}(C) := \{f: X \rightarrow Y : X \text{ and } Y \text{ are retracts of each other}\}$.

Moreover Iso and W_{end} are incomparable in general, $\text{Iso} \vee W_{\text{end}} = W_{\text{obj}}$, $W_{\text{obj}} \subseteq W_{\text{ret}} \subseteq W_{\text{cyc}}$, and $W_{\text{ret}} = W_{\text{obj}}$ in every category whose endomorphism monoids are finite.

Proof. Functors preserve identities, isomorphisms, isomorphic pairs of objects, backs, endomorphisms, automorphisms and retractions, so all classes are natural once they are samenesses. W_{obj} , W_{ret} : Lemma 5.2, “mutual retract” being an equivalence relation containing $\cong$ (a retract of a retract is a retract). W_{end} : composites of endomorphisms of X are endomorphisms; if gf and f are endomorphisms of X then $g: X \rightarrow X$; if gf and g are endomorphisms then $\text{dom } f = \text{cod } gf = \text{cod } g = \text{dom } g = \text{cod } f$. Aut is an intersection. Incomparability: in the category with two objects and one isomorphism between them, $\text{Iso} \not\subseteq W_{\text{end}}$; in the monoid $\{1, t, t^2\}$ with $t^3 = t^2$, $W_{\text{end}} \not\subseteq \text{Iso}$. Join: W_{obj} is a sameness containing both; and if $f: X \rightarrow Y$ with an isomorphism $h: Y \rightarrow X$, then $f = h^{-1} \circ (hf)$ with $hf \in W_{\text{end}}$, $h^{-1} \in \text{Iso}$, so f lies in the join. If X is a retract of Y via $s: X \rightarrow Y$, $r: Y \rightarrow X$, $rs = \text{id}$, then $e \mapsto ser$ is an injection $\text{End}(X) \rightarrow \text{End}(Y)$ (with left inverse $u \mapsto rus$); with finite endomorphism monoids, mutual retracts give a bijection, so $\text{id}_Y = ser$ for some e , and s is a split epimorphism as well as a split monomorphism, hence an isomorphism. $\square$

Theorem 5.4 (the cut lemma). *Let W be a natural sameness. If for some category C the class $W(C)$ contains a morphism $f: X \rightarrow Y$ with $Y \not\preceq X$, then $W = \top$. Hence every natural sameness other than Mor is contained in W_{cyc} , and W_{cyc} is the largest proper natural sameness. Dually, if $W(C)$ contains a morphism $f: X \rightarrow Y$ with $X \neq Y$, then $W \supseteq \text{Iso}$.*

Proof. Let $U := \{Z : Z \preceq X\}$ and $V := \text{Ob } C \setminus U$; then $X \in U$, $Y \in V$, and no morphism goes from V to U (if $Z' \rightarrow Z$ with $Z \preceq X$ then $Z' \preceq X$). So $U \mapsto 0$, $V \mapsto 1$ defines a functor $F: C \rightarrow [1]$ to the arrow category, with $F(f)$ the non-identity arrow u of $[1]$. Naturality gives $u \in W([1])$, and every morphism of every category is the image of u under a functor $[1] \rightarrow D$; so $W = \top$. For the second statement let I be the category with two objects $0, 1$ and exactly one morphism between any two objects (the walking isomorphism). Any assignment of objects of C to $\{0, 1\}$ extends uniquely to a functor $C \rightarrow I$; choose $X \mapsto 0$, $Y \mapsto 1$, so $F(f)$ is the isomorphism $u: 0 \rightarrow 1$ and $u \in W(I)$. An isomorphism $h: A \rightarrow B$ of any D

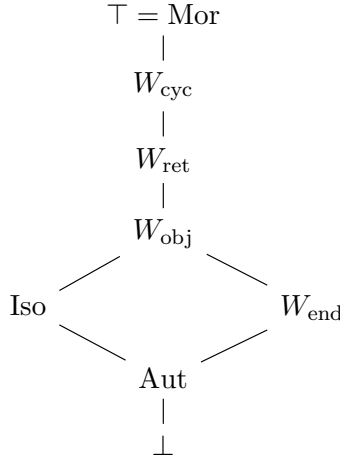


FIGURE 3. Natural samenesses of Proposition 5.3; Iso and W_{end} are incomparable with meet Aut and join W_{obj} . Further natural samenesses exist between W_{obj} and W_{cyc} (Remark 5.5).

Alt text: A Hasse diagram: bottom, then automorphisms, then two incomparable elements (isomorphisms; endomorphisms), their join (source isomorphic to target), then mutual retracts, then the cycle class, then the top.

is the image of u under the functor $I \rightarrow D$ sending $0 \mapsto A$, $1 \mapsto B$, $u \mapsto h$, $u^{-1} \mapsto h^{-1}$ (also when $A = B$). So $W \supseteq \text{Iso}$. $\square$

Remark 5.5 (towards a classification). By Lemma 5.2, natural equivalence relations on objects give natural samenesses W_R ; by the cut lemma every such relation other than the total one is contained in mutual reachability. There are infinitely many: the transitive closure of any natural symmetric relation defined by a finite pattern (mutual retracts; pairs $u: X \rightarrow Y$, $v: Y \rightarrow X$ with $uvu = u$ and $vuv = v$; ...) is one. Not every natural sameness is of this form (W_{end} is not), and two-out-of-three couples the object part with the endomorphism part ($\text{Iso} \vee W_{\text{end}} = W_{\text{obj}}$). We do not know the complete list. Restricting a natural sameness $W \subseteq W_{\text{end}}$ to one-object categories yields a natural assignment $M \mapsto N(M)$ of two-out-of-three-closed submonoids of monoids, of which we know only $\{1\}$, the group of units, and M ; whether this restriction determines W — i.e. whether membership of an endomorphism in $W(C)$ depends only on its image in $\text{End}_C(X)$ — is a further classification question.

6. LOCALIZATION VERSUS REFLECTION IN A TOPOS

Let $\mathcal{E}$ be a topos. Every object X has a *support* $\text{supp} X \rightarrow 1$, the image of $X \rightarrow 1$; the subterminal objects (subobjects of 1) form a reflective full

subcategory $\text{Sub}(1)$ with reflector $X \mapsto \text{supp}X$, the (-1) -truncation [5, A1.3]. Put $W_{\text{supp}} := \{f: X \rightarrow Y : \text{supp}X = \text{supp}Y\}$, the class inverted by this reflector.

Theorem 6.1. *In a topos $\mathcal{E}$:*

- (1) $\text{Loc}(W_{\text{cyc}}) = \text{Sub}(1)$, *the subterminal objects;*
- (2) $\text{Sat}(W_{\text{cyc}}) = W_{\text{supp}}$;
- (3) $W_{\text{cyc}} \subseteq W_{\text{supp}}$, *with equality iff any two objects with the same support admit morphisms in both directions.*

Proof. (1) Let Z be subterminal. For any Y , $\text{Hom}(Y, Z)$ is empty or a singleton, and it is non-empty iff $\text{supp}Y \leq Z$ in $\text{Sub}(1)$ (a map $Y \rightarrow Z$ factors through the image of $Y \rightarrow 1$, and conversely). If $f: X \rightarrow Y$ has a back g , then $\text{supp}X \leq \text{supp}Y$ because of f and $\text{supp}Y \leq \text{supp}X$ because of g ; so $\text{Hom}(Y, Z) = \text{Hom}(X, Z)$ and Z is W_{cyc} -local. Conversely let Z be W_{cyc} -local and X arbitrary. The fold map $\nabla: X \sqcup X \rightarrow X$ has the back j_1 , so $\nabla \in W_{\text{cyc}}$ and $\text{Hom}(X, Z) \cong \text{Hom}(X \sqcup X, Z) = \text{Hom}(X, Z) \times \text{Hom}(X, Z)$ via the diagonal; a set in bijection with its square through the diagonal has at most one element. So $\text{Hom}(X, Z)$ has at most one element for every X , i.e. $Z \rightarrow 1$ is a monomorphism: Z is subterminal. (2) W_{supp} is the class inverted by the reflector onto $\text{Sub}(1)$, hence is saturated with local objects $\text{Sub}(1)$; by (1) it has the same local objects as W_{cyc} , so it is $\text{Sat}(W_{\text{cyc}})$. (3) The inclusion was shown in (1). If $f: X \rightarrow Y$ lies in W_{supp} and objects with equal support admit maps both ways, then f has a back; the converse is the definition. $\square$

$$\begin{array}{ccc}
 \mathcal{E} & \xrightarrow{\gamma} & \mathcal{E}[W_{\text{cyc}}^{-1}] = (\text{Ob}\mathcal{E}, \preceq) \\
 \text{supp} \downarrow & \swarrow \text{dashed} & \\
 \text{Sub}(1) & &
 \end{array}$$

FIGURE 4. Localization versus reflection at W_{cyc} in a topos: the localization is the hom-preorder (Corollary 4.4); the reflection onto the local objects is the support, i.e. the (-1) -truncation (Theorem 6.1); the dashed comparison is an equivalence iff W_{cyc} is saturated.

Alt text: A triangle: the topos maps to its hom-preorder by the localization and to the subterminal objects by the support; a dashed arrow compares the hom-preorder with the subterminals.

Example 6.2 (sets and simplicial sets). In **Set** and in **sSet** the supports are $\emptyset$ and 1 , and two non-empty objects admit maps in both directions (in **sSet** through a vertex, $Y \rightarrow \Delta^0 \rightarrow X$). So $W_{\text{cyc}} = W_{\text{supp}}$ is saturated, it

is the class of maps $f: X \rightarrow Y$ with ($X = \emptyset$ iff $Y = \emptyset$), i.e. the (-1) -truncated equivalences, and $\mathcal{E}[W_{\text{cyc}}^{-1}] \simeq \text{Sub}(1) = \{\emptyset < 1\}$. For simplicial sets this class is the join of the weak homotopy equivalences with the 0-coskeletal equivalences [9, Thm. 6.2, $n = -1$]: the antisymmetry sameness of simplicial sets is the (-1) -truncation.

Example 6.3 (graphs: W_{cyc} is not saturated). In **Gph** let X be the path $v_0 \rightarrow v_1$ and Y the graph with one vertex and one loop (the terminal graph). Both have support 1, and there is a map $X \rightarrow Y$; but $\text{Hom}(Y, X) = \emptyset$, since a loop must map to a loop. So $X \rightarrow Y$ lies in $W_{\text{supp}} \setminus W_{\text{cyc}}$: W_{cyc} is strictly smaller than its saturation, the localization **Gph** $[W_{\text{cyc}}^{-1}]$ (the homomorphism preorder, Corollary 4.5) is strictly finer than the reflection **Gph** $\rightarrow$ **Sub**(1), and the comparison of Figure 4 identifies all graphs with at least one edge.

The pattern is the one observed in [10] for joins of Grothendieck topologies: the sameness lattice sees a relation — here reachability, there a pro-limit — that the saturated, reflective level forgets.

7. CONCLUDING REMARKS

The cycle class W_{cyc} answers the question “antisymmetric up to which sameness?” with a canonical element of $\text{Same}(C)$, whose extremes classify acyclic, EI and strongly connected categories, whose localization is the EI-reflection, and which under binary coproducts is the preorder reflection in disguise. Two questions remain. First, the classification of natural samenesses (Remark 5.5): is every natural sameness generated by natural equivalence relations on objects and natural two-out-of-three-closed submonoids of endomorphism monoids? Second, for which categories without binary coproducts is the EI-reflection still a preorder — the free category on $x \rightleftarrows y$ shows that some hypothesis is needed, and finite categories with a zero object satisfy the conclusion by the same fold-map argument applied to the zero morphisms.

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